Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/23/2007
PI Name:	Lizee, Greg Ph.D.		
Project Title:	Tumor Antigen Cross-presentation by Dendritic Cells Cultured in Modeled Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	77054	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
Start Date:	01/01/2006	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to FSB as of FY2006, per program changes at that time, per J. Cohen/HQ and JSC/J. Dardano and previous information fr D. Risin/JSC in Mar 2006 when PI was C. Savary (jvp 6/2009) NOTE: Received NCE to 9/30/2008 (from 11/30/2007) per P. Denkins/JSC (5/07); title changed to "Tumor Antigen Cross-presentation by Dendritic Cells Cultured in Modeled Microgravity" from "Study of Molecular Lesions in Dendritic Cells Cultivated in Modeled Microgravity" NOTE: Dr. Lizee assumed project in January 2006 (9/06 HQ info); for information prior to this date, see Savary (PI)		
Key Personnel Changes/Previous PI:	Greg Lizee replaced Cheryln Savary as PI in early 2006. Title changed to "Tumor Antigen Cross-presentation by Dendritic Cells Cultured in Modeled Microgravity" from "Study of Molecular Lesions in Dendritic Cells Cultivated in Modeled Microgravity."		
COI Name (Institution):			
Grant/Contract No.:	NNJ04HB05G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space flight has a significant negative impact on immune function. This is a concern because microorganisms that cause life-threatening opportunistic infections on earth thrive aboard spacecraft. We have found that human dendritic cell (DC) maturation and functions were attenuated in a microgravity analog culture, i.e., the rotary cell culture system (RCCS). DCs are considered to be the most efficient of antigen-presenting cells. Because they play a central role in development of innate and acquired immune responses, a defect in DC would be expected to result in a broad impairment of immune function, not unlike that observed in astronauts during or after space flight. Yet there is nothing known about the effects of space flight on DC. In our previous studies we observed that, when compared to DC generated in standard static culture conditions, DC grown in RCCS displayed a decreased ability to phagocytose A. fumigatus conidia, reduced levels of surface MHC class II and CD80 costimulatory molecules, and impaired production of IL-12 after exposure to the fungus. These observations suggest that DC generated in modeled microgravity lack the processing and presentation properties needed to mount an effective antifungal immune response. If a similar situation occurs in true microgravity, it could be an underlying factor contributing to the immune blunting associated with space flight, and would emphasize the need to develop effective countermeasures to protect the health of astronauts. Before countermeasures can be developed, though, we must identify targets where countermeasures could be effective. Therefore, the objectives of this study are to identify the cellular and molecular lesions in DC-genesis and function that occur in an in vitro microgravity analog. We plan to: (a) identify molecular lesions in human DC maturation and function that occur in the modeled microgravity environment of the RCCS; (b) examine DC maturation and function in the murine RCCS model; and (c) evaluate the effects of dietary nucleotides on murine and human DC generation and function in the RCCS. There is overwhelming data from space flight experiments to suggest that a defect in antigen presenting cells might occur in microgravity. Until routine testing of space travelers is possible, we must make use of ground-based models proven to mimic (to the degree that it is possible on earth) some of the effects on immune performance known to be associated with space flight. The RCCS is such a model. In fact, in our experience with this model, we have already detected several defects in DC maturation and function that could explain some of the immune blunting reported for astronauts/cosmonauts. A novel aspect of our studies has been the inclusion of Aspergillus and Candida antigens as targets for immune testing. We are particularly interested in gaining more information on host defense against Aspergillus because of the urgent need to develop new treatments for this opportunistic mould that causes life-threatening infections in immunocompromised patients. These fungi colonize spacecraft and could also impact the health of astronauts. Thus, our studies address a significant problem in space medicine. In our quest to identify lesions associated with impaired DC maturation and function in modeled microgravity, we will contribute new knowledge to the area of DC immunobiology in general. Moreover, we should be able to identify targets for development of countermeasures to buffer against these negative effects to the benefit of both astronauts and earth-bound patients.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Greg Lizee replaced Cheryln Savary as PI in early 2006. Title changed to "Tumor Antigen Cross-presentation by Dendritic Cells Cultured in Modeled Microgravity" from "Study of Molecular Lesions in Dendritic Cells Cultivated in Modeled Microgravity." See Savary for FY2006 report on this project.
Bibliography Type:	Description: (Last Updated:)